

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2025

Course: B.Tech (AIDS and Allied)

Semester : IV

Subject Code & Name: Data Analysis (BTAIC401)

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

- 1 Each question carries 12 marks.
- 2 Question No. 1 will be compulsory and include objective-type questions.
- 3 Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- 4 The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- 5 Use of non-programmable scientific calculators is allowed.
- 6 Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- 1 Which of the following represents a type of bias in statistics?
A) Confidence Interval B) Hypothesis Testing
C) Sampling Bias D) Degrees of Freedom
1
- 2 Non-parametric analysis is used when:
A) Data follows a normal distribution
B) Data violates assumptions of parametric tests
C) Sample size is infinite D) Data is free of bias
1
- 3 Measures of central tendency include:
A) Mean, Median, Mode B) Variance and Range
C) Chi-square Test D) Regression Coefficient
1
- 4 Which of the following is a measure of spread?
A) Variance B) Mean C) Mode D) Proportion
1
- 5 Freeman-Tukey transformations include which of the following?
A) Square root and arcsine transformations
B) Log and exponential transformations
C) Chi-square and F-tests D) Variance stabilization
1
- 6 The logit transformation is commonly applied to:
A) Categorical data B) Proportion or probability data
C) Skewed financial data D) Normally distributed data
1
- 7 The Chi-square test is used for:
A) Testing the mean of a single sample B) Testing variance of a population
C) Goodness of fit or association in categorical data
D) Regression analysis
1

- 8 Which of the following tests is non-parametric? 1
 A) T-test B) Z-test C) Wilcoxon rank-sum test D) ANOVA
- 9 McNemar's test is applied to: 1
 A) Comparing means of two groups
 B) Paired nominal data in contingency tables
 C) Regression coefficients D) Homogeneity of variance
- 10 One-way ANOVA is used to: 1
 A) Compare variances of two groups
 B) Compare means across multiple groups
 C) Analyze categorical data D) Measure correlation
- 11 Kruskal-Wallis test is: 1
 A) A parametric ANOVA test B) A test for proportions
 C) A non-parametric alternative to ANOVA D) A goodness of fit test
- 12 ANCOVA adjusts for: 1
 A) Means only B) Variances only
 C) Covariates and reduces error variance D) Regression slopes

Q. 2 Solve the following. 12

- A)** Explain hypothesis testing with its types of errors. How can power and robustness improve the reliability of results? 6
- B)** Two teaching methods are compared to evaluate student performance.
 • Method A: Sample size = 50, Mean score = 75, Standard deviation = 10
 • Method B: Sample size = 60, Mean score = 78, Standard deviation = 12 6
- At 5% significance level, test whether there is a significant difference in the average scores.

Q.3 Solve the following. 12

- A)** Explain the need for log and exponential transforms with suitable examples. 6
- B)** Discuss the role of data transformation and standardization in improving statistical inference. 6

Q. 4 Solve Any Two of the following. 12

- A)** Discuss the Wilcoxon rank-sum/Mann-Whitney U test. In which scenarios is it preferred? 6
- B)** Discuss Fisher's exact test and its application in small sample sizes. 6
- C)** Given the following data: [10, 15, 25, 40, 80, 160] 6

Apply a log transformation and compute the transformed values. Explain why log transformation is useful in this scenario.

Q.5 Solve Any Two of the following.

12

- A)** Differentiate between Z-test and T-test. When is each test applied? Explain the F-test for comparing variances. Provide an example. **6**
- B)** Explain the concept and importance of measures of association in contingency tables. Describe McNemar's test. How does it differ from the Chi-square test? **6**
- C)** A battery manufacturer claims that the average life of their batteries is 100 hours. A random sample of 25 batteries shows a mean life of 96 hours, with a known population standard deviation of 10 hours. Test the manufacturer's claim at a 5% significance level using a Z-test. **6**

Q. 6 Solve Any Two of the following.

12

- A)** Explain one-way ANOVA with an example. What assumptions are involved? Discuss two-way ANOVA and its applications in analyzing experimental data. **6**
- B)** Explain ANCOVA and how it adjusts for covariates in data analysis. Compare Kruskal-Wallis ANOVA and Friedman ANOVA test. In which situations are they used? **6**
- C)** A researcher wants to compare the average test scores of students from **three different classrooms** to see if there is a significant difference in their performance. The test scores are:

Classroom A	Classroom B	Classroom C
65	70	68
60	75	72
62	72	70

6

At a **5% significance level**, test whether there is a significant difference between the mean scores of the three classrooms using One-Way ANOVA.

***** End *****